LOURENZ O. BALIBER, MSc

Researcher | Biophysics | Data Scientist
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Results-driven Physics postgraduate with expertise in soft matter and polymers, biophysics, end-to-end machine learning pipelines, predictive modeling, and data-driven application deployment. Proficient in Python, R, and MATLAB with hands-on experience in data analysis and scientific computing. Experienced collaborating with international research teams.

TECHNICAL SKILLS

- Programming & Data Tools: **Python (NumPy, Pandas, SciPy, Matplotlib, Scikit-learn, PyTorch), R, MATLAB, Mathematica, C++, Git/Version Control, Linux/Unix & Shell Scripting**
- Data Science & ML: **Supervised & unsupervised learning, regression and classification, time-series analysis, statistical modeling, computer vision (OpenCV), feature engineering, cross-validation & model evaluation**
- Deployment & Web: **Flask, Streamlit, React, REST APIs, end-to-end ML deployment, SQL, MongoDB, automated data pipelines (n8n, Zapier), JavaScript/HTML/CSS**
- Scientific Computing: **Molecular dynamics simulation (LAMMPS, GROMACS), physics-based simulation, passive microrheology, particle tracking, fractal dimension analysis**
- Other: **Tableau, LaTeX, scientific writing, CAD modeling**

EXPERIENCE

Exchange Student Researcher (Physical Chemistry) *2025 – 2026*
Tokyo University of Marine Science and Technology
- Processed and analyzed **2,000+ fluorescence microscopy image frames** using custom Mathematica pipelines, reducing manual analysis time by **~60%**.
- Developed image analysis workflows correlating structural heterogeneity with particle dynamics from passive Brownian motion tracking across **40+ samples**.
- Applied statistical modeling to characterize phase-separated biopolymer network formation; results and data analysis contributed to a **peer-reviewed publication**.

Graduate Researcher – MSc Thesis *2024 – 2026*
Medical Biophysics Group, University of San Carlos
- Performed quantitative analysis of fluorescence microscopy and passive particle tracking on datasets of **~8,000 data points** for mixed carrageenan hydrogel systems.
- Implemented data processing pipelines in Python/MATLAB for MSD, viscoelastic moduli extraction, and gel network characterization across **50+ particle trajectories**.
- Collaborated with international researchers on biophysics and data analysis methodology.

Research Assistant *Aug – Nov 2023*
DOST PCIEERD NICER Project — 3D-BTM: Bio-fabrication of Bone, Muscle & Pancreatic Tissue Models for Printable and Injectable Scaffolds
- Supported data collection and analysis for biomimetic tissue engineering experiments, processing **15+ experimental datasets**.
- Assisted in processing and interpreting biomechanical measurement datasets; contributed to a **government-funded national research project**.

Research Intern *Jun – Jul 2023*
Medical Biophysics Group, University of San Carlos
- Conducted optical tweezers and microrheology experiments on complex fluids.

- Processed experimental data to quantify local viscoelastic properties of biological samples, achieving reproducibility within **±3% measurement error**.

EDUCATION

MSc – Physics *2024 – 2026*
University of San Carlos, Cebu City, Philippines
- Thesis: "Gelation Mechanism and Network Formation of FITC-Labeled Mixed Kappa/Lambda Carrageenan Gels"
- DOST ASTHRDP Scholar

BSc – Applied Physics *2020 – 2024*
University of San Carlos, Cebu City, Philippines
- Thesis: "Characterization of Fibrin-Carrageenan Mixtures as a Hydrogel Model Using Passive Particle Tracking and AFM for Skeletal Muscle Tissue Engineering"
- DOST JLSS Scholar (2022–2024) | Ironwood Scholar (2020–2024) | Dean's Lister (2020–2022)

PUBLICATIONS

- Baliber, L. et al. (2026). Brownian motion in a phase-separated biopolymer network. *Proceedings of the 44th Samahang Pisika ng Pilipinas Physics Conference*, SPP-2026-PA-01.
- Baliber, L., et al. (2025). Surface roughness and fractal dimension characterization of the topography in pristine and defected nanostructure arrays. Proceedings of the Samahang Pisika ng Pilipinas, 43, SPP-2025-2G-02.
- Baliber, L., et al. (2024). Passive microrheology in carrageenan–fibrin mixtures using video microscopy. Proceedings of the Samahang Pisika ng Pilipinas, 42, SPP-2024-1E-05.

PROJECTS

Satellite Launch & Orbit Simulator — Python Orbital Mechanics Model *2025*
- Built a **2D physics simulation modeling** a satellite's trajectory from launch to orbital insertion at LEO, solving the equations of motion via custom RK4 integration under gravitational and atmospheric drag forces while **accounting for fuel consumption and mass loss.**
- Visualized trajectory, velocity, and altitude profiles in real time using **Matplotlib animation**, validating simulation output against analytical orbital mechanics solutions.

Stroke Detection System — End-to-End ML Pipeline *2024*
https://github.com/fitaness12345/Stroke-Dataset-Prediction
- Built a complete stroke detection system from **EDA to web deployment** on a dataset of **5000+ patient records**.
- Deployed as an interactive web application using **Flask + Streamlit**, enabling real-time risk prediction with visual feature importance output.

Balibot — Discord Bot *2023*
github.com/fitaness12345/balibot
- Developed and deployed **Balibot** using **discord.py**, serving **200+ users across 5+ servers**.
- Features include automated moderation, slash commands, API integrations, and **96.3% uptime** over 6 months.

HONORS & AWARDS

- 25th SPVM National Physics Conference – Poster Presenter (2023)
- 33rd Annual Fr. Heinrich Schoenig Biosymposium – Poster Presenter (2024)

- ODU REYES Python & Subatomic Physics Mentorship (2021)
- Top 3 Scholar, USC DOST Thesis Grants (2024)
- 44th SPP Conference – Poster Presenter (2026)

AFFILIATIONS

- Creative Writer - The Josenian Premier (2019 - 2020)
- **Administrative Positions**
  - Internal Affairs Councilor - Carolinian Physics Society (2021 - 2022)
  - President – USC Physics and Astronomy Society (2022 - 2023)
  - Councilor - School of Arts and Sciences Council (2022 - 2023)
  - Physics Representative – DOST ASTHRDP Officer (2024-2025)
- **Co-organized Conferences**
  - 2nd USC-VSU Biophysics Symposium (2023)
  - 8th MBG Biophysics Symposium (2024)
  - 3rd USC-VSU Biophysics Symposium (2025)
  - 8th International and 16th Annual National Scientific Meeting of the Philippine Society for Cell Biology (2025)
  - 18th International Hydrocolloids Conference (2026)